

457.568 Statistical Hydrology, Fall 2022

R Code Training Assignment #1 (Probabilistic Modeling)

Due before the lecture of Oct-18, 2022

<Notice> Include responses to every question below, and make sure your final assignment is well organized (i.e., numbered responses, full sentences, code attached at the end). This makes it easier to grade & to give partial credits as well.

1. (R Loop Exercise) You are about to print the following message using for loops in R.

```
print(paste("The year is", 2010))  
"The year is 2010"  
print(paste("The year is", 2011))  
"The year is 2011"  
print(paste("The year is", 2012))  
"The year is 2012"  
print(paste("The year is", 2013))  
"The year is 2013"  
print(paste("The year is", 2014))  
"The year is 2014"  
print(paste("The year is", 2015))  
"The year is 2015"
```

- A. Using a for loop, create a command that produces the same results with the above example.
- B. Again, use two for loops to create a command that produces the results with both month and the year. Assume that there are only three months: "Jan", "Feb", and "Mar." Your results should start from Jan-2010 and finish with Mar-2015.
2. 윤진아씨는 빵집을 운영하고 있다. 하루 평균 20명의 방문자가 방문하고, 매일 방문하는 방문자의 수는 Poisson 분포를 따른다고 가정한다 또한, 방문자는 재고가 없지 않는 이상 케이크를 무조건 구매하며, 하루 최소 방문자는 1명, 최대 방문자의 수는 40명이라고 가정한다. 케이크 하나의 가격은 35,000원이고, 만드는데 드는 비용은 25,000원이다. 윤진아씨는 혼자 빵집을 운영하기 때문에 영업 전에 미리 케이크를 준비해 놓으며, 당일에 팔지 못하는 케이크는 폐기한다고 한다.
- A. 매일 케이크를 얼마나 준비하는 것이 기대 이윤 (즉, 총 수입과 총 비용의 차이)을 극대화할 수 있는 방안인가?

- B. 손님이 방문하였으나 케이크 재고가 부족하여 팔지 못하면 대신 기념품을 제공하고, 해당 기념품의 제작 비용은 20,000원이라고 하자. 기대 이윤을 극대화하기 위해 준비해야 하는 케이크의 수는 어떻게 바뀌는가?
3. Conduct the probabilistic model analysis using the data provided as “OwegoCreekFlow-1.csv” following the steps below.
- A. Load the data into R using the `read.csv()` function. Then, use the `aggregate()` function to average the monthly flow data for each calendar month, generating 12 monthly average values. Create a figure of this monthly hydrograph. Determine and report which month has the lowest average monthly flow. This will be the ‘design month’: by examining water demand and supply conflicts during the driest month, our analysis will consider the most constraining time of year for water withdrawals.
- B. Create a new variable in R (‘owego.min.month’) that only contains the data for the design month. This variable should be a vector that has 48 elements, one for each year of the record.
- C. Plot a histogram of the data in `owego.min.month` using the `hist()` function. Briefly comment on the shape of the distribution. Is it symmetric? Is it skewed?
- D. Determine an appropriate distribution with which to model the monthly flows for the design month. Fit the normal, log-normal, and gamma distributions to these data using maximum likelihood. You can use functions in R designed for MLE, such as `fitdistr()` in the MASS package.
- E. Calculate and report the log likelihood using the MLE parameter estimates for each distribution. This is included in the output of the `fitdistr()` function. Report the fitted parameters and log-likelihood value for each model.
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- F. Calculate and report the AIC for each model. Which distribution will you choose to model these data?
- G. The AIC values above give a sense of which distribution of the three (normal, lognormal, and gamma) fits the data the best relative to the other distributions, but it doesn’t tell you if the fit is good in an absolute sense. For the model that you choose in F, create a P-P plot to check the fit directly. Comment on the goodness-of-fit given this plot.
- H. To triple-check the fit of your chosen distribution, use a chi-squared test to conduct a hypothesis test. You can use the `chisq.test()` function. To conduct this test, first calculate the observed number of observations between (0-50 cfs), (50-100 cfs), and (>100 cfs). An easy way to do this is to combine the table and the cut functions:

```
obs.breaks <- cut(owego_min_month, breaks=c(0,50,100,Inf))  
obs.counts <- table(obs.breaks)
```

`cut()` assigns each observation to a particular bin, while `table()` counts the total number of times observations fall into each of the three bins. Run the variables `obs.breaks` and

obs.counts in the console to see this for yourselves.

In addition to the observed counts, you need to calculate the expected probabilities (exp.probs) that the data would fall into each bin, assuming the distribution you chose was a good fit to the data (hint: to do this, you can make use of the cdf function for your chosen distribution, e.g., pgamma() if you chose the gamma distribution). Note that the chisq.test() takes expected probabilities, not expected counts, although exp.probs should be a vector of the same length as the vector obs.counts. Report the X-squared statistic, degrees of freedom, and the p-value from the chi-square test returned by chisq.test(x=obs.counts, p=exp.probs). How do you interpret the results of this test?

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R Code Training Assignment #2 (Extreme Event Analysis)

Due before the lecture of Nov-15, 2022

<Notice> Include responses to every question below, and make sure your final assignment is well organized (i.e., numbered responses, full sentences, code attached at the end). This makes it easier to grade & to give partial credits as well.

1. Conduct the extreme event analysis using the data provided as “fallcreek.csv” following the steps below.
 - A. Load the data into R using the read.csv() function into a variable in R called FallCreek. Open the “fallcreek.csv” file to see how the data is organized. Save the flow of the “fallcreek.csv” file into a variable named ‘AnnMaxFlow.’
 - B. It is always helpful to visualize the annual maxima as a time series against water year. However, to do this, we first need to manipulate the original data, specifically the dates. Using the mutate verb in the dplyr package, change the Date column to be a Date object in R (hint: you can use the as.Date() function within the mutate verb).
 - C. Using the ggplot2 package, create a line graph between the date and the annual maxima, as well as two histograms of the annual maxima, one on the original data scale and another on the log10 scale. Comment on the shape of the distribution on both scales.
 - D. We are going to examine the fit of a GEV distribution to the annual maxima using the “ismev” library in R. After installing and loading the library, you can fit a GEV distribution to your data as follows:

my.gev.fit <- gev.fit(FallCreek\$AnnMaxFlow)

The MLE parameter estimates can be recovered with my.gev.fit\$mle. Report your estimated parameters (location, scale, shape).

- E. Using ggplot, examine how well the GEV fits the data using a probability plot. To do this, use dplyr and mutate to add two new columns to FallCreek: plotting.pos and gev.cdf.
 - i. plotting.pos will be the plotting positions, and can be calculated using the rank() function on the AnnMaxFlow data
 - ii. gev.cdf will be the cdf of the fitted GEV evaluated on your data, and can be calculated using the GEV cdf function gevf() in the ismev package. There are two arguments to the function: a vector of 3 parameters values, and a vector of data to pass through the cdf.

Create a probability plot of plotting positions against model-based cdf values. Include a 1:1 line (Google how for ggplot). Comment on how well the GEV distribution fits your data?

- F. Develop point estimates of the following return period events: 2-year, 10-year, 25-year,

50-year, 100-year, 200-year. To do this:

- i. Create a new data.frame FallCreekDesignEvents with a single column named return.period” equal to the vector of return levels listed above.
 - ii. Using dplyr and mutate, create a new column (“exceed.prob”) of exceedance probabilities.
 - iii. Use mutate to create another column (“return.level”) of the design event values. Here, you can use your fitted parameter estimates and the GEV quantile function, which is conveniently provided by the gevq() function. The gevq() function takes two arguments: a vector of 3 parameter values, and a vector of exceedance probabilities.
- G. Report a table of return periods and return levels. Using ggplot, develop a return period plot using these design event estimates. Make sure that the x-axis is on a log10 scale and connect your estimates with a line.

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2. In this exercise we are going to revisit the calculation of Fall Creek design floods based on annual maxima, but this time we’ll compare results from the GEV and Log-Pearson Type 3 (LP3) distributions. In the later part of the exercise, you will use the Generalized Pareto distribution (GPD) for your analysis.
- A. First, add onto the return period plot from 1-G in a different color estimates of design events using a log Pearson type III distribution.
- i. You can find a frequency factor table as an attachment. Feel free to round your skewness estimate to the nearest tenths place.
 - ii. Use the mutate verb to add the LP3-based return levels to the FallCreekDesignEvents data frame before plotting.
- B. Add the empirical return periods against the annual maxima on the same plot.
- i. Empirical return periods can be derived from empirical non-exceedance probabilities, which were stored in the FallCreek data frame in Problem 1 to create a P-P plot. (In other words, $\text{Emp.return.period} = 1/(1-\text{plotting.pos})$)
- C. In ggplot(), a slightly different syntax is needed if you want to plot data from multiple data frames (e.g., FallCreekDesignEvents and FallCreek) on the same plot. In the example below, two different lines are added from data in one data frame (called df1), and points are added from a different data frame (called df2). The color argument in the aesthetics is used to add labels to the legend for each plotting feature.

```
ggplot() +  
  geom_line(data=df1, aes(x = , y = , color = “myline1”)) +  
  geom_line(data=df1, aes(x = , y = , color = “myline2”)) +  
  geom_point(data=df2, aes(x = , y = , color = “mypoints”))
```

Make sure all of the elements on your plot are clearly labeled in the legend. You can use the same label for the upper and lower bounds of the GEV return levels. Comment on how similar or different the return level estimates are between the GEV and LP3 models, and how the observed data compares to the models.